OpenMagnetics

Steps

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Tool menu

# Magnetic Summary

MAS Exports

Core Exports

Coil Exports

Circuit Simulators Exports

## Core data

### Core Shape

| | |
|---|---|
| Name | T 64/36/25 |
| No. Stacks | 1 |
| Effective length | 0.16 m |
| Effective area | 353.19 $mm^2$ |
| Effective volume | 55029.1 $mm^3$ |
| Minimum Area | 353.19 $mm^2$ |

### Core Gapping

### Material Parameters

| | | |
|---|---|---|
| Name | N87 | |
| Manufacturer | TDK | |
| | 25°C | 100°C |
| Permeance (AL value) | 6.56 $\mu H/(tu.)^2$ | 11.35 $\mu H/(tu.)^2$ |
| Initial Permeability ($\mu_i$) | 2304 | 3983 |
| Eff. Permeability ($\mu_e$) | 2303 | 3983 |
| Resistivity | 10 Ωm | 10 Ωm |
| Curie Temperature | 210 °C | |
| Density | 4.85 $Mg/m^3$ | |

## Coil data

### Coil Global Parameters

### Primary

| | |
|---|---|
| No. turns | 4 |
| No. parallels | 1 |
| Wire | Litz Round 3.15 - Grade 1 |

### Secondary

| | |
|---|---|
| No. turns | 35 |
| No. parallels | 1 |
| Wire | Litz undefined |

## Simulation result per Operating Point

### Op. Point No. 1

### Core

| | |
|---|---|
| Mag. Ind. | 181.5 µH |
| Core losses | 1.73 W |
| Core temp. | 110.5 °C |
| B peak | 85.60 mT |

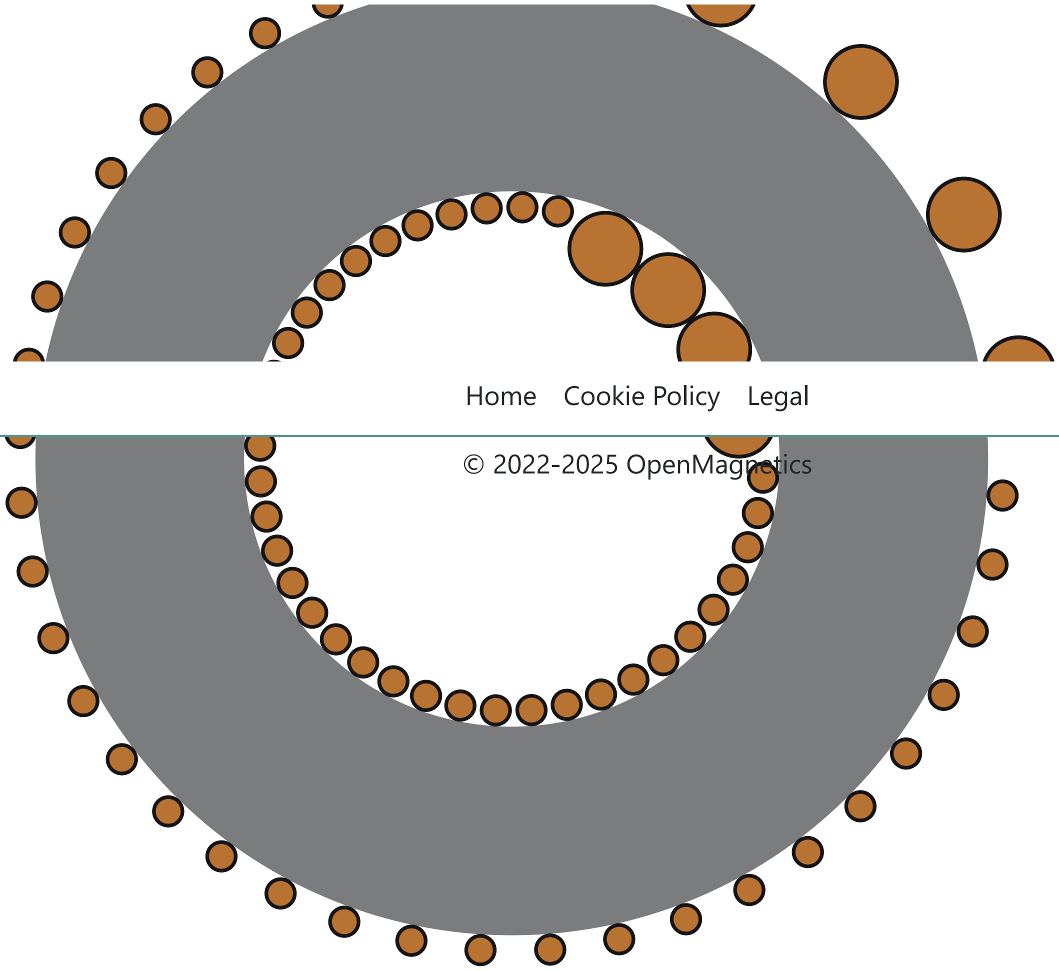
### Coil

| Winding | DC Res. | Curr. Density | Wind. Loss | Leak. Ind. |
|---|---|---|---|---|
| Primary | 1.07 mΩ | 15.18 A/mm² | 16.1 W | |
| Secondary | 62.47 mΩ | 10.29 A/mm² | 8.60 W | 379 nH |

### Coil Losses Breakdown

| Winding | Ohmic Loss | Skin Loss | Prox. Loss |
|---|---|---|---|
| Primary | 15.16 W | 1.19 mW | 0.94 W |
| Secondary | 7.44 W | 1.58 mW | 1.15 W |

### Core Coil



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